Exp. No: 3

Force of Attraction between Two Electric Charges and Two Parallel Conductors Separated by Varying Distance

Aim:

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

Objectives:

1. To compute the electrostatic force of attraction or repulsion between two point charges using Coulomb's law.

2. To compute the magnetic force per unit length between two current-carrying parallel conductors.

3. To analyze and plot the variation of force with separation distance and charge or current magnitude.

Program:

Test case (i): Determination and plotting of force between two electric charges with varying distance

```
clc;

clear;

q1 = input('Enter the charge q1=');

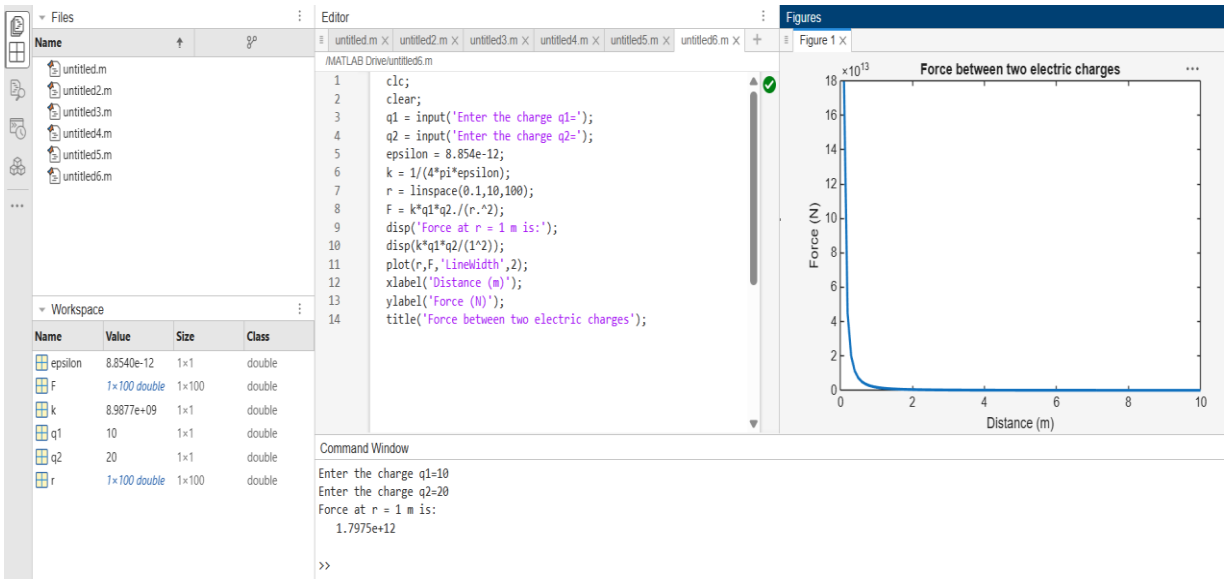
q2 = input('Enter the charge q2=');

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);
```

```
r = linspace(0.1,10,100);
F = k*q1*q2./(r.^2);
disp('Force at r = 1 m is:');
disp(k*q1*q2/(1^2));
plot(r,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force (N)');
title('Force between two electric charges');
```

output:



Test case (ii)

Determination and plotting of force between two electric charges with different magnitudes

Code:

```
clc;
q1 = 5e-6;
q2 = 2e-6;
```

```
epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.5,5,100);

F = k*q1*q2./(r.^2);

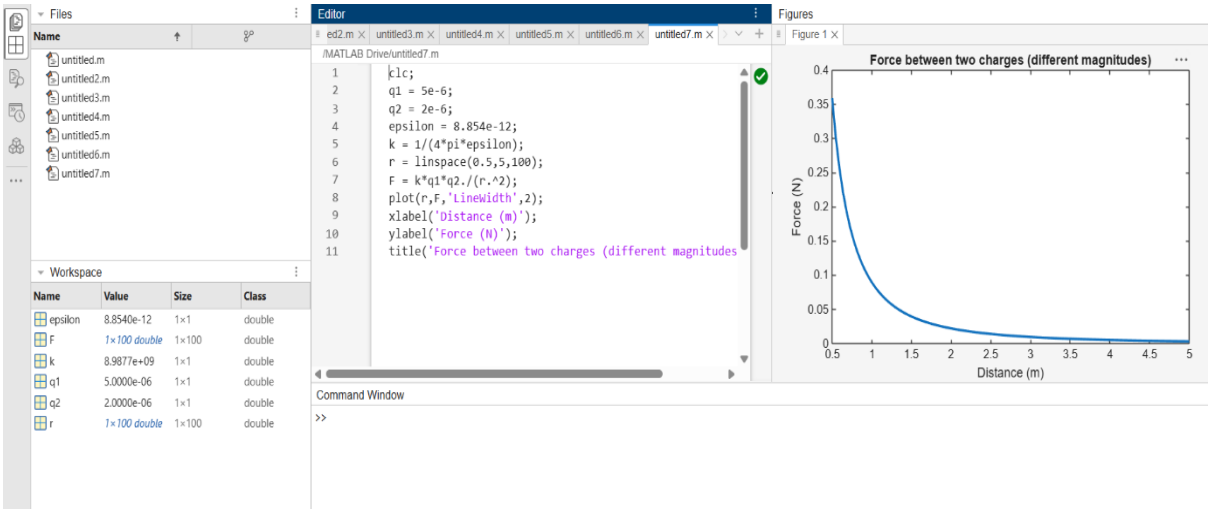
plot(r,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force (N)');

title('Force between two charges (different magnitudes)');
```

output:



Test case (iii)

Determination and plotting of force per unit length between two parallel conductors with varying distance

Code:

```
clc;

mu0 = 4*pi*1e-7;

I1 = 5;
```

I2 = 10;

d = linspace(0.1,10,100);

F = (mu0*I1*I2)./(2*pi*d);

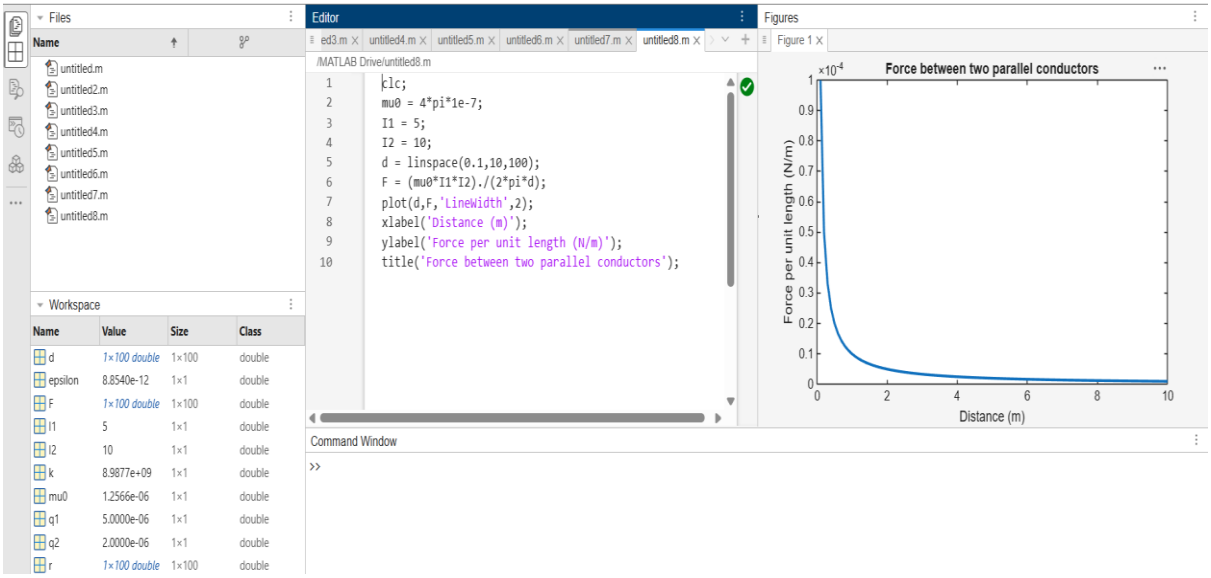
plot(d,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force per unit length (N/m)');

title('Force between two parallel conductors');

output:



Test case (iv):

Determination and plotting of force between two parallel conductors carrying equal currents

Code:

clc;

mu0 = 4*pi*1e-7;

I1 = 3;

I2 = 3;

d = linspace(0.1,2,100);

```
F = (mu0*I1*I2)./(2*pi*d);

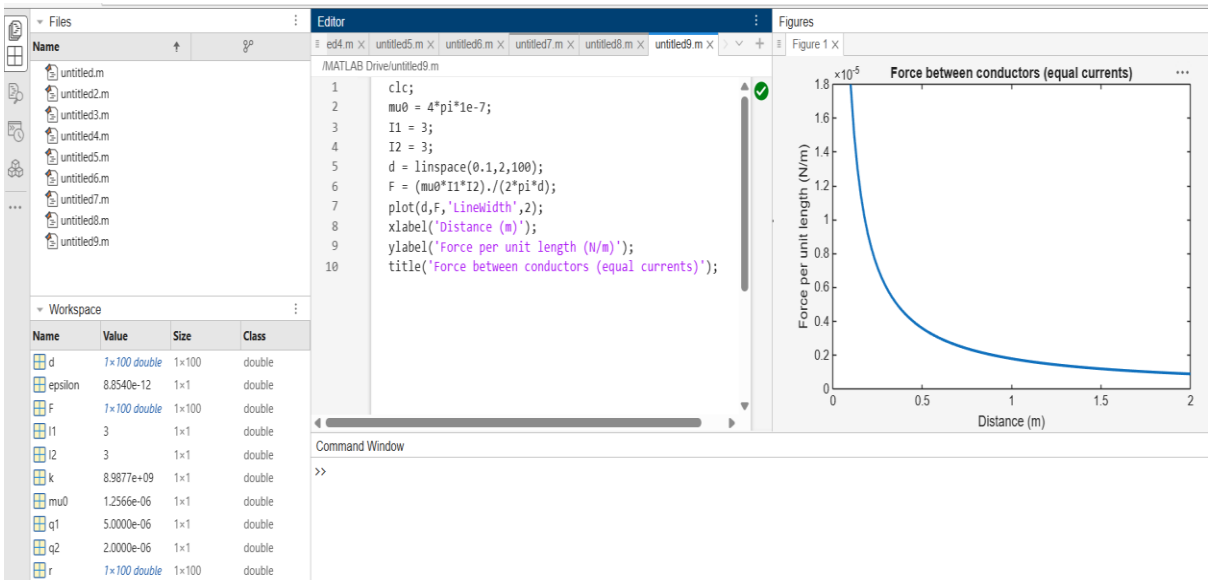
plot(d,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force per unit length (N/m)');

title('Force between conductors (equal currents)');
```

output:



Rubrics:

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightag (%)	Marks Given	Marks Obtained
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifies most quantities and models correctly.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25	2.5	
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel current force calculations correctly with units.	Calculations are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25	2.5	
3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends.and checks theoretical distance dependence.	Trend and plots are correct with minor omissions.	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30	3	
4	Explain and report results	Explain and report results	Explains attraction/repulsion and reports assumptions, units and conclusions clearly.	Conclusion is correct with minor documentation gaps.	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20	2	
Total							**100**	**10**	